

Automated Driver Drowsiness Detection System

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Abstract— In recent years, road safety has become an increasingly pressing issue due to an increase in the number of accidents all over the world. Accidents are a major threat to road safety, and fatigue and drowsiness are the Primary causes of such incidents. In order to prevent such accidents, it is essential to detect and aware of signs that cause drowsiness or fatigue in a driver, and issue a timely warning. A range of technologies have been created to detect drowsy driving; however, computer vision and image processing technology can efficiently detect the drowsiness using facial expressions. As described a system has been proposed that continuously monitors the driver's facial expression by analyzing the eyes and mouth movement in order to detect any signs of drowsiness or emotional shifts. If any such changes are detected, the system will alert the driver and helps in avoiding potential danger. This system provides a low-cost and resource-efficient approach to monitoring driver behavior and it utilizes facial Landmark analysis to assess a driver's performance without the need for additional sensors or equipment.

Keywords— Road-Safety, Driver Drowsiness, Fatigue, Facial Expressions, Computer Vision, Image Processing, Alert.

I. INTRODUCTION

Road accidents in India continue to be a major cause of death and disability, accounting for 11% of all road-related deaths in the world. In 2019, there were 449,002 accidents resulting in 151,113 deaths and 451,361 injuries, a slight decrease from 2018[16]. The implementation of several measures that focus on road safety, is thought to have contributed to the decline. National Highways accounted for 35.7% of deaths, State Highways 24.8%, and other roads 39%. The majority of deaths (85%) were in the 18-60 age group. These accidents are often caused by an unrecognized factor - drowsy driving.

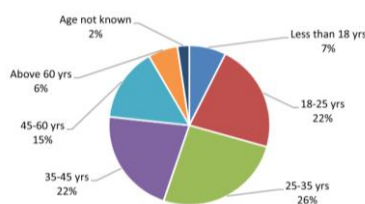


Fig. 1. Age profile of road accident deaths in 2019

What is Drowsy Driving?

Drowsy driving is when a driver operates a vehicle while being excessively tired or fatigued. It is a major safety concern on the roads, as it can lead to a lack of concentration and reaction time, as well as potentially dangerous driving behaviours. The effects of drowsy driving can be similar to those of drunk driving, often resulting in slower reaction times and impaired judgement[21]. Drivers who are feeling drowsy are more likely to make mistakes and fail to recognize potential hazards on the road, making them more susceptible to causing a crash. In addition to the potential for a fatal crash, drowsy driving can also result in costly property damage, as well as a variety of other serious injuries to both the driver and other people.

According to the National Highway Traffic Safety Administration, drowsy driving is responsible for about 1 in 5 fatal crashes. The AAA Foundation for Traffic Safety estimates that drowsy driving resulted in 72,000 crashes, 44,000 injuries, and 800 deaths in 2013. The National Sleep Foundation reports that 60% of drivers have driven while drowsy in the past year, and 37% of drivers have fallen asleep at the wheel. It is estimated that drowsy driving costs the U.S. economy \$109 billion each year.

In order to avoid drowsy driving and ensure driver safety, it is important for drivers to take the necessary precautions. This includes getting a good night's sleep before driving, taking regular breaks, and avoiding driving late at night or early in the morning. Additionally, drivers should be aware of the warning signs of drowsy driving, such as yawning, heavy eyelids, and difficulty keeping their eyes open [14].

Drivers can also take advantage of detection systems that are designed to alert them if they are feeling drowsy. These systems use sensors and artificial intelligence algorithms to detect signs of fatigue and alert the driver before anything happens. These technologies that are designed to detect signs of drowsiness in drivers, such as yawning, blinking, and erratic steering, and alert them so they can take corrective action. These systems use a variety of sensors to monitor a driver's behaviour, such as video cameras and infrared sensors [18], and use artificial intelligence algorithms to detect signs of fatigue. If the system detects signs of drowsiness, it will alert the driver with a warning tone or vibration, as well as display a visual alert helping the drivers to reduce the risk of drowsy driving-related crashes.

In this paper, we explore the various models and methods used for drowsiness detection and propose the most suitable system. We evaluated different approaches to detect drowsiness, and found machine learning based approach with computer vision provide accurate results.

Our work compares various machine learning algorithms for detecting drowsiness and evaluate the effectiveness of each algorithm [8]. To determine which one would be best suited for detecting drowsiness. The algorithms we evaluated include Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree (DT), Naïve bayes, Random Forest and Logistic Regression. We analyse the performance of each algorithm in terms of accuracy and used it for training and testing.

The best trained classifier is used in the process of real-time detection. The system includes a video camera that will capture the live video of the driver's behaviour. The captured video is split into individual frames. These frames are processed and used for facial landmarking. OpenCV and dlib can be used together to process images. The dlib tool kit is used to provides algorithms for facial recognition, facial Landmark detection, and face morphing [5]. It also provides a wide range of image processing functions, such as image filtering, image segmentation, and image registration. OpenCV can be used to detect objects in an image and dlib can be used to perform facial recognition. Additionally, OpenCV and dlib can be used together to perform image segmentation, image registration, and image filtering[17].

These images are used detect signs of drowsiness in drivers, such as yawning, blinking and to help drivers stay alert and safe on the road. This system does not include any additional sensors or external devices for detecting the driver's drowsiness.

The structure of this paper is organized as follows. In Section II, we defined various methods related to our work in detail. In Section III, we provide the methodology that we had worked on to develop a cost-effective and efficient model. In Section IV, the Experimental results have been provided for the proposed system. In Section V, the conclusion has described and finally, in Section VI, the future enhancement has been discussed.

II. RELATED WORKS

There are a variety of detection systems that can help ensure safe driving. These include:

Lane Departure Warning Systems: Y. Saito *et al.* [6] proposed a driver assistance system with a dual control scheme for identifying driver drowsiness and preventing lane departure accidents. This system uses cameras and sensors to detect when a vehicle is drifting out of its lane and alert the driver with a warning tone or vibration.

Blind Spot Detection Systems: G. Liu *et al.* [9] proposed a radar-based blind spot detection and warning system for driver assistance. This system uses sensors to detect when another vehicle is in a driver's blind spot and alert the driver with a warning tone or vibration.

Adaptive Cruise Control Systems: C. Han *et al.* [7] proposed a paper on robust cruise control of heterogeneous connected vehicle systems. This system provides low latency wireless vehicle-to-vehicle communication.

Respiratory Signal Analysis Systems: F. Guede-Fernández *et al.* [4] proposed a driver drowsiness detection system based on respiratory signal analysis. This technique uses sensors to measure a driver's respiratory rate, which is then used to assess the level of drowsiness. The sensors detect changes in the driver's breathing rate and analyse them to identify signs of drowsiness.

Drowsiness Detection System Based on EEG Signals: M. Pathak *et al.* [15] designed a single channel EEG acquisition system for detection of drowsiness which aims to design an economical and portable single-channel EEG acquisition system for drowsiness detection.

PERCLOS Method: K. S. Sankaran *et al.* [18] proposed a system that focuses on eye closure by an effective method named Percentage eye Closure (PERCLOS). The PERCLOS establishes a parameter level to detect the drowsiness. This method uses infrared or visible light sensors to measure the angle of the eyes and compare it to a predefined baseline.

Hybrid Machine Learning Algorithm: A. Altameem *et al.* [1] developed a hybrid machine learning algorithm for drowsiness detection that combines computer vision, artificial intelligence, and deep learning to detect signs of fatigue under different lighting situations. The advantages of using this approach for drowsiness detection such as its ability to detect fatigue earlier than traditional methods

III. METHODOLOGY

The proposed methodology is divided into two sections training a model and Real time monitoring.

A. Training a Model:

Training a model is the process of creating a machine learning algorithm that can accurately predict outcomes. In this we use various techniques and algorithms to create the model and optimize its performance.

The process consists of the following steps:

1) Collection of Data:

To develop an effective driver drowsiness detection model, we must first collect data on the various indicators of drowsiness, such as eye-closing, yawning, eye-opening, no-yawning and other physical signs of fatigue. Here we take a dataset containing 5216 images of 640 X 480 pixels. The images were taken under various conditions.

2) Splitting the Data:

Once the data is collected, the data should then be split into two sets: training data and testing data. In this model we split the data into 9:1 ratio where 90% of the data will be training data which is used to train the model, and 10% of the data will be testing data which is used to assess the performance of the model. Finally, the performance of the model is then evaluated to measure the accuracy of the model's predictions.

Yawning and eye closure are both common signs of drowsiness, so we included these types of data in our dataset. They include a range of different images with both eyes open and closed, as well as images with and without yawning. The figure 2 to figure 5 denotes the sample of the data used to train and test the model.



Fig. 2. Example for Yawning data



Fig. 3. Example for No- yawning data

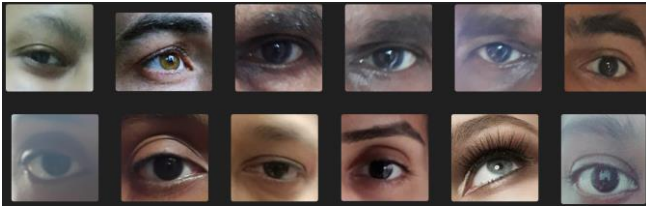


Fig. 4. Example for opened-eye data



Fig. 5. Example for closed-eye data

3) Pre-Processing the Data:

In this process, the data undergoes image pre-processing, feature extraction, skin-segmentation, noise reduction and feature selection. The images are resized and cropped to 64 X 64 pixels to have a consistent format and involves identifying and extracting the features. These steps are performed to ensure that the data is clean, consistent, and suitable for the machine learning model.

4) Model Evaluation:

We used several algorithms to detect drowsiness, including K-Nearest Neighbors (KNN), Naive Bayes, Support Vector Machines (SVMs), Random Forests, Logistic Regression, and Decision Trees. Each algorithm was evaluated based on its ability to accurately detect drowsiness.

K-Nearest Neighbors (KNN): KNN is a non-parametric, lazy learning algorithm that can be used for both classification and regression. KNN can be effective in detecting drowsiness since it takes into account the distance between points to determine the class of a given point. KNN can be used to generate a model that can detect drowsiness with a reasonable degree of accuracy. We trained and tested the model with the data and able to get 80.821%.

Naive Bayes: Naive Bayes is a probabilistic machine learning algorithm that uses Bayes' theorem for predicting the probability of a given class. Naive Bayes can be used to detect drowsiness by taking into account the probability of a given input being drowsy or not. By using Naive Bayes, we had achieved 69.18% accuracy.

Modified SVMs: Support Vector Machines are a supervised learning algorithm that can be used for both classification and regression. SVMs are effective in detecting drowsiness since they can accurately separate the data points belonging to different classes. Modified SVMs are a variation of regular SVMs that are optimized for specific tasks. Modified SVMs are especially useful in detecting drowsiness since they can be trained to better recognize patterns associated with drowsiness. Modified SVMs can be used to generate a model that can detect drowsiness with a higher degree of accuracy than regular SVMs. The SVM classifier is trained to classify images as drowsy or non-drowsy based on the training data. Modified SVM produce an accuracy of 95.20% making it best suitable algorithm for detecting drowsiness.

Decision Trees(DTs): Decision Trees are a supervised machine learning algorithm that can be used for both classification and regression. Decision trees can be used to detect drowsiness since they can accurately separate the data points belonging to different classes. Decision Trees are used to generate a model that can detect drowsiness with a 88.70% of accuracy.

Random Forests: Random Forests are an ensemble learning algorithm that combines multiple decision trees and outputs the most frequent class. They can provide an accuracy of 92.46% for the provided data.

Logistic Regression: Logistic Regression is a supervised machine learning algorithm that can be used for both binary and multiclass classification. Logistic regression can be used to detect drowsiness since it can classify the data points into different classes based on the input features. Logistic regression was able to produce 93.49% accuracy.

In conclusion, After evaluating several algorithms for the task of detecting drowsiness, it was found that the modified Support Vector Machines (SVMs) algorithm had the highest accuracy at 95.20%. These results suggest that Modified SVMs are the most suitable algorithm for detecting drowsiness.

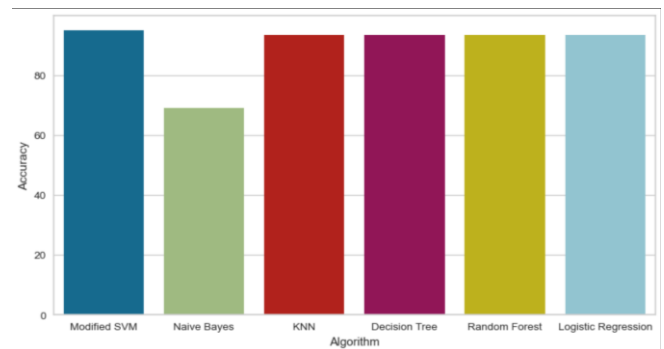


Fig. 6. Accuracy of each algorithm

B. Real-Time Monitoring

In Real-time monitoring we continuously monitor a driver's alertness and alert them if they show signs of drowsiness while driving. We use a variety of methods to detect drowsiness, such as monitoring eye movements, analysing the driver's facial expressions or monitoring the driver's yawning. If drowsiness is detected, the device can alert the driver through auditory or visual cues to help them stay awake and alert. Real-time monitoring can be an

effective way to help reduce the risk of accidents caused by drowsy driving.

The whole system is shown in Figure 7. It represents the proposed system's framework.

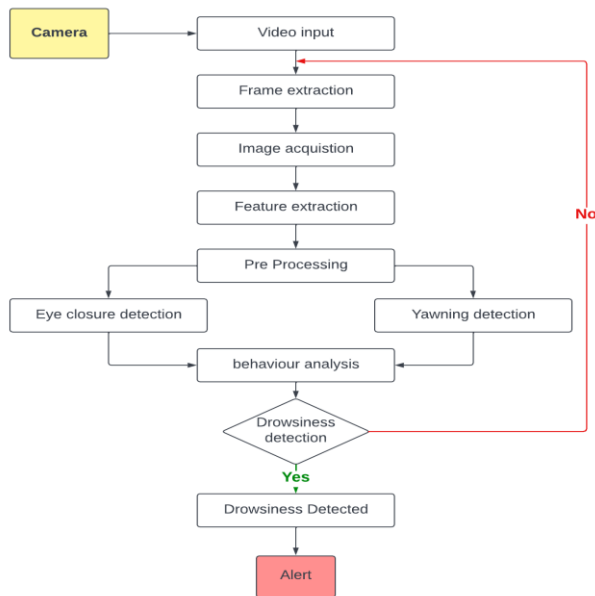


Fig. 7. System Architecture

1)Video Input:

This can be done using video cameras or other video capture devices, which can be mounted in the vehicle or on the driver's body.

In the proposed system, the video input was captured using a webcam, which is a type of digital video camera that is typically connected to a computer or other device through a USB port. The webcam was used to capture a streaming video of the driver, which was then processed in real-time to detect signs of drowsiness and it allows continuous monitoring of the driver's alertness levels.

2)Frame Extraction:

Frame Extraction refers to the process of dividing a video stream into individual frames or images, which can then be processed individually.

In this case, the video stream is captured at a rate of 20 frames per second (fps), then each frame represents a snapshot of the driver's appearance at that moment in time. By dividing the video stream into individual frames at a rate of 20 fps, the system can analyze each frame separately to detect signs of drowsiness, such as yawning and blinking.

3)Features Extraction:

Feature extraction is a process used in machine learning and image processing to extract important features or characteristics from data.

To extract features from the images, the system would first need to detect the location of these features in the video frames. This can be done using techniques such as template matching or feature-based object detection. Once the location of the eyes and mouth has been detected, dlib can

be used to extract specific features from these regions, such as the size and shape of the eyes or the movement of the mouth. These features are used as input to machine learning algorithms that can analyze the data to detect signs of drowsiness, such as yawning and blinking. This approach to feature extraction can be effective in helping to improve the accuracy of drowsiness detection systems.

4)Eye Closure Detection:

In this system, the Eye Aspect Ratio (EAR) is used to detect eye closure in real-time video streams. It works by tracking the distances between certain points on the eye, such as the outer corner of the eye and the inner corner of the eye (vertically) and the distance between the outer corner of the eye and the center of the eye (horizontally). If the ratio of these distances falls below a certain threshold, this can be taken as an indication that the eye is closing, which can be tracked over time to detect when the eye is fully closed and alert the driver if necessary.

5)Yawning Detection:

We use the Mouth Aspect Ratio (MAR) as a measure to detect drowsy driving by tracking the Aspect of the mouth in a video stream. The MAR is calculated as the ratio of the distance between the outer corners of the mouth and the distance between the top and bottom lips. When the mouth is open, the MAR is relatively high, but as the mouth closes, the MAR decreases. By tracking the MAR over time, it is possible to detect when the mouth is closing and when it is fully closed.

The system would need to track the location of certain points on the mouth in each frame of the video. Once the locations of the relevant points on the mouth have been determined, the MAR ratio can be calculated for each frame. If the MAR ratio falls below a certain threshold, this can be taken as an indication that the driver is drowsy.

6)Drowsiness Detection:

By tracking the EAR and MAR over time, the system able to detect when the eye is closing and the mouth is closing. The EAR and MAR ratios are calculated for each frame. If either ratio falls below a certain threshold, this can be taken as an indication that the driver is drowsy.

Immediately, the system will display a visual or audio alert. The visual alert is a message displayed on a screen, while the audio alert is a beep. By alerting the driver to their drowsy state, the system can help prevent accidents caused by drowsy driving.

IV. EXPERIMENTAL RESULTS

The experiment conducted to detect drowsiness using Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) results the information about driver's eyes were closed or opened, yawning and number of times the driver yawned.

Here are some examples of the outcomes produced by our proposed system. Figures 8, 9, and 10 depict the various states of driver alertness detected by the system.



Fig. 8. Eyes open and Mouth closed



Fig. 9. Eyes closed and Yawning

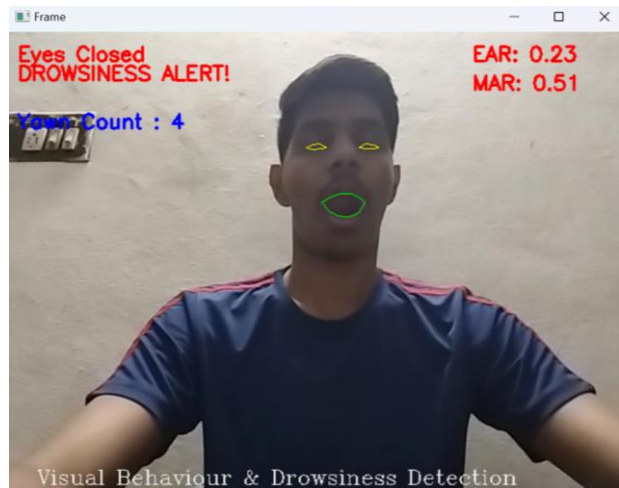


Fig. 10. Eyes closed and Yawning count

V. CONCLUSION

In this study, we investigated the effectiveness of real-time drowsiness detection techniques in various lighting conditions. We implemented and compared multiple machine learning algorithms and image processing clustering techniques, and found that the modified support

vector machine (SVM) algorithm had the highest accuracy at 95.20%. Our system was able to detect drowsy driving by analyzing the driver's facial expressions and triggering an alarm when the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) fell below a certain threshold. The proposed algorithm performed well at an optimal distance from the light source, but its efficiency decreased with increasing distance and decreasing light. This system has the advantage of not requiring any special devices or sensors to be worn by the driver and can be implemented using a camera or any other imaging system mounted in the vehicle.

VI. FUTURE ENHANCEMENT

The system was designed with a focus on real-time monitoring, and emphasized flexibility, adaptability, and the ability to be enhanced. there are several potential areas for future enhancement of the system.

One possibility is to develop the system as a standalone product that can be installed in an automobile to monitor the driver. this would allow the system to be used in a variety of different vehicles and would not require the driver to have a separate device. Another option is to implement the system as a smartphone application that can be installed on a driver's smartphone. the driver could start the application and place the smartphone in a position where the camera is focused on their face, allowing the system to monitor their facial expressions and detect signs of drowsy driving. This approach would be convenient and portable, as the driver would only need to have their smartphone with them to use the system.

They have the potential to significantly improve the effectiveness and convenience of the system for detecting drowsy driving.

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